

1. Fundamentals & Big Data

What is Big Data? What are its characteristics (the Vs)?

Big Data refers to datasets so large or complex that traditional data processing systems cannot handle them efficiently. The key characteristics are known as the 5 Vs: Volume (large amounts of data), Velocity (high speed of generation), Variety (different data types and formats), Veracity (uncertainty or quality of data), and Value (usefulness derived from data).

How does Hadoop help address Big Data challenges?

Hadoop provides a distributed storage and processing framework. It allows data to be stored across many inexpensive servers (HDFS) and processed in parallel (MapReduce), improving scalability, fault tolerance, and cost efficiency.

What are the main components of Hadoop (HDFS, MapReduce, YARN)?

HDFS stores data across clusters, MapReduce processes the data in parallel, and YARN manages cluster resources and job scheduling.

What is the difference between Hadoop 1.x and Hadoop 2.x (YARN introduction)?

Hadoop 1.x used a single JobTracker for both resource management and job scheduling, causing scalability issues. Hadoop 2.x introduced YARN (Yet Another Resource Negotiator), separating these functions and enabling multiple processing frameworks beyond MapReduce.

2. HDFS (Hadoop Distributed File System)

What is HDFS?

HDFS is Hadoop's distributed file system that stores large files by splitting them into blocks and distributing them across multiple nodes in a cluster.

What is a block? What is the default block size?

A block is the smallest unit of data storage in HDFS. The default block size is typically 128 MB (previously 64 MB in older versions).

What is replication factor?

Replication factor defines how many copies of each block are stored across the cluster for fault tolerance. The default is 3.

What happens when a DataNode fails?

The NameNode detects the failure via heartbeats and re-replicates the missing blocks from other nodes to maintain the desired replication factor.

Explain NameNode, Secondary NameNode, JournalNode.

NameNode manages metadata and the namespace of HDFS. Secondary NameNode periodically merges edit logs with the fsimage to keep backups updated. JournalNodes are used in High Availability mode to store edit logs for shared access between Active and Standby NameNodes.

What is a checkpoint?

A checkpoint is a merged copy of the NameNode's edit log and fsimage, reducing recovery time after restarts.

Explain rack awareness.

Rack awareness is the policy used by HDFS to distribute replicas across different racks for better fault tolerance and network performance.

What commands do you use to interact with HDFS?

Examples: `hdfs dfs -ls`, `-put`, `-get`, `-du`, `-rm`, `-mv`, `-cat`, etc.

What is fsck in HDFS?

fsck (File System Check) verifies the integrity of files in HDFS and reports any missing or corrupt blocks.

3. MapReduce

What is MapReduce?

MapReduce is a programming model for processing large datasets in parallel across a distributed cluster. It splits input data into independent chunks processed by map tasks, and the results are aggregated by reduce tasks.

Describe the phases: Map, Shuffle & Sort, Reduce.

Map: transforms input data into key-value pairs. Shuffle & Sort: groups values by keys and transfers them to reducers. Reduce: processes each group of values to produce the final output.

What is a Combiner?

A Combiner is an optional mini-reducer that performs local aggregation of mapper output to reduce data transfer during the shuffle phase.

What is partitioning?

Partitioning determines which reducer a key-value pair is sent to. By default, Hadoop uses hash partitioning based on the key.

What is the role of InputFormat and OutputFormat?

InputFormat defines how input data is split and read (e.g., TextInputFormat), while OutputFormat defines how results are written (e.g., TextOutputFormat).

How does the framework handle failures of mappers or reducers?

Hadoop automatically re-executes failed tasks on other nodes to ensure fault tolerance.

What is speculative execution?

Speculative execution runs duplicate tasks on slow nodes to finish the job faster by using the earliest result.

When would you set number of reducers to 0?

When the job requires only map outputs (no aggregation), such as filtering or transformations that don't need reducing.

4. YARN & Resource Management

What is YARN?

YARN (Yet Another Resource Negotiator) is the resource management layer in Hadoop 2.x that allows multiple applications to share cluster resources efficiently.

Explain ResourceManager, NodeManager, ApplicationMaster.

ResourceManager allocates cluster resources, NodeManager runs tasks on nodes and monitors resource usage, and ApplicationMaster manages the execution of a specific application.

How does scheduling work (capacity scheduler, FIFO, Fair)?

FIFO executes jobs in submission order, Fair Scheduler shares resources equally among jobs, and Capacity Scheduler divides resources by configured capacities per queue.

What happens when AM fails?

If an ApplicationMaster fails, YARN restarts it automatically on another node up to a configured retry limit.

5. Ecosystem Tools (Hive, Pig, HBase, Sqoop, etc.)

What is Hive? What are internal vs external tables?

Hive is a data warehouse tool built on Hadoop for querying and analyzing data using SQL-like language (HiveQL). Internal tables store data in Hive's warehouse directory and delete it on drop, while external tables reference external HDFS data and preserve it when dropped.

What is SerDe in Hive?

SerDe stands for Serializer/Deserializer. It converts data between Hadoop's internal format and readable formats (e.g., CSV, JSON).

What is Pig? Describe its architecture.

Pig is a high-level platform for processing data using a scripting language called Pig Latin. Its architecture includes a parser, optimizer, and execution engine that converts scripts into MapReduce jobs.

What is HBase?

HBase is a NoSQL, column-oriented database that runs on top of HDFS and allows real-time read/write access to large datasets.

How is HBase different from traditional RDBMS?

HBase is schema-less, horizontally scalable, and optimized for sparse data, while RDBMS is schema-based, vertically scaled, and supports complex queries.

What is Sqoop? Use cases for it.

Sqoop is a tool for transferring data between Hadoop and relational databases (import/export). Commonly used for ETL processes.

What is Flume or Kafka (for ingestion)?

Flume and Kafka are tools for collecting and streaming data into Hadoop in real time. Flume is event-based; Kafka is a distributed messaging system for large-scale data pipelines.

6. Advanced / Scenario-Based

How do you optimize a MapReduce job (tuning, memory, combiners)?

Use combiners, adjust mapper/reducer memory settings, compress intermediate data, and optimize input formats. Also tune the number of mappers and reducers and enable speculative execution where beneficial.

How to deal with data skew?

Use custom partitioners or sampling to distribute load evenly across reducers. Skewed data can slow down reducers handling heavy keys.

Describe join strategies in Hadoop (map-side join, reduce-side join).

Map-side joins happen before the shuffle phase when both datasets are partitioned and sorted identically. Reduce-side joins send both datasets to reducers keyed by join fields, suitable for large and unaligned datasets.

How to migrate data between clusters?

Use DistCp (Distributed Copy) to copy data between Hadoop clusters efficiently over parallel mappers.

Scenario: You have many small files in HDFS — what issues arise? How to resolve?

Small files cause excessive metadata load on NameNode. Solutions: combine files using SequenceFile, HAR (Hadoop Archive), or store them in HBase.

Scenario: You need to update a record in HDFS — how do you handle that?

HDFS is append-only and doesn't support in-place updates. Use HBase or overwrite the file with a new version.

How would you monitor cluster health & job performance?

Use tools like Ambari, Cloudera Manager, Ganglia, or built-in web UIs for monitoring resource usage, node status, and job progress.